Mateo Robles

Creative Technologist | Generative Art, WebGL, Interactive AI
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Education

Santa Clara University Santa Clara, CA
B.S. Computer Science, Data Science Emphasis *June 2026*

- Relevant coursework: Artificial Intelligence, Machine Learning, Algorithms, Programming Languages

Experience

SyberLabs Bay Area
Founder and Software Engineer, open-source research and engineering studio *2026 – Present*

- Built and open-sourced 4 products in 2026 (RISE, OmniOS, Relay, Dynamic Circulator) by setting product direction for all four, architecting three, and co-developing with a second engineer through code review and CI.
- Launched **RISE** at rise.syberlabs.io, an audiovisual reader that paces text against museum imagery and Web Audio, with a user-controlled photosensitivity toggle for rhythmic effects.
- Sought design critique on **RISE** from four HCI, typography and media-studies experts, and turned it into an ongoing redesign of reading pace, type scale and image-text pairing.
- Cut the **RISE** JavaScript bundle by **15.4 MB**, holding first load to about **59 KB** Brotli, by moving reading content into a SHA-256-verified data plane in a Vite app backed by Cloudflare Workers, PostgreSQL and Redis.
- Made every LLM answer traceable to its inputs by building **OmniOS** (in development), a local-first Next.js/TypeScript canvas where personas see only the live data wired into them (Polymarket, World Bank, OpenAlex, Hacker News), running keyless on local models via Ollama.
- Let OmniOS users add unfamiliar APIs without hand-written integrations by building a compiler that turns OpenAPI specs and **MCP** tool schemas into typed blocks, with approval-gated writes and idempotent execution that reports when a write's outcome is uncertain.
- Modeled adaptive crowd behavior in Dynamic Circulator with decaying preference memory and UCB edge learning, served through a **FastAPI** REST/WebSocket API to a React/Vite dashboard and covered by **100+ pytest** tests.

Projects

Engines | *Generative art series* | sykosyber.github.io/engines 2025 – 2026

- Created **Engines**, seven seeded, reproducible generative-art engines in WebGL2 and Canvas 2D (RK4-integrated strange attractors, phase-interference fields, space-colonization growth), each with live palette and parameter controls.

SkyPredict V2 | *Machine learning course project, team of 3* | github.com/SyberLabs/skypredict-v2 2026

- Predicted U.S. flight delays before departure at **0.713 ROC-AUC**, 0.349 PR-AUC and **19.23-minute MAE** on a 1.16M-flight chronological holdout by building a leakage-aware LightGBM pipeline over **6.96M** BTS records.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS; familiar with C++, C, Go, C#
Creative Coding: WebGL2, Canvas 2D, Web Audio, generative and procedural systems, React, Vite
AI / LLM: LLM applications, AI agents, tool calling, Model Context Protocol (MCP), OpenAPI, Ollama, Anthropic and Gemini APIs, inference logging
Backend: FastAPI, Pydantic, REST, WebSockets, PostgreSQL, Redis, Cloudflare Workers, D1, Drizzle ORM
Testing / Infra: Git, GitHub Actions CI/CD, pytest, Vitest, Playwright, ESLint
ML / Data: PyTorch, scikit-learn, LightGBM, NumPy, SciPy, pandas, feature engineering, time-series, model evaluation